Ram Calculus and Augmented Elliptic Inverses on Degenerating Ricci-Flat Trees

Residuals, finite-rank changes, complex lengths, and the path-selection boundary

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Abstract

This report examines the pinned OpenAI manuscript on path selection and an elliptic inequality for a fixed finite tree of four-dimensional Ricci-flat spaces. We reconstruct its weighted augmented inverse, including every angular channel, finite core sources and tests, critical seam data, partial disconnections, quadratic complex-length domains, and the all-order scalar expansion used in the final contradiction. From the actual private Ram Calculus passages we derive a full-operator residual framework, a low/high trace-block elimination with transported residuals, finite-rank source/test update bounds, and a uniform nonlinear and holomorphic-parameter criterion. A conditional observable perturbation estimate identifies the accuracy needed to preserve the sequential conclusion. These are bounded-operator refinements, not a finite-matrix proof of noncompact Fredholmness or a new proof of the source's path-selection theorem. Exact algebraic and scalar model checks are separated from the unperformed continuum validation. Applicability is L5 because the methods meet the source's principal analytic construction directly. Contribution assessment: C2/5, explicit source-specific residual and update criteria provide a modest improvement in auditability, but no validated geometric inverse constants, stronger tree inequality, or independent global proof is obtained.

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1 Scope, evidence, and claim types

The primary object is *Path selection and an elliptic inequality on degenerating Ricci-flat trees*, September 24, 2026, at OpenAI commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

We call it S [1]. Its seven substantive sections were read in full, not merely its abstract or catalogue entry. There are no separate substantive appendices in its main input tree. The preamble, main file, bibliography, and scale diagram were also inspected. Source labels in this report refer to that version. The local source directory is an export without Git metadata: a failed `rev-parse` is not evidence of a different version or an independent attestation of the pin. A separately supplied source-audit record compared all eleven build inputs, totaling 3208 lines, byte-for-byte with public raw files at the pin and reported eleven matches. This report retains that provenance record and rechecks its local input hashes; it does not describe the author's failed Git query as a pin verification.

The private inputs are the actual passages of Core-5 [2], nonlinear Report 1 [3], and PDE-first Report 8 [4]. Their finite-dimensional formulas are not silently promoted to infinite-dimensional theorems. We prove the bounded-operator statements below, and state the analytic hypotheses needed for their use. No private text or private filenames were used as web queries. The public check used public titles and URLs only.

Type	Meaning in this report
Imported	S's geometric inverse, nonlinear family, tail expansion, scalar admissibility, and path-selection claims, with their proof dependencies explained. They are not credited as new results here.
Proved locally	The typed residual bounds, trace Schur residual identity, finite-rank update criterion, nonlinear radius theorem, and observable error estimate under their displayed assumptions. Most underlying algebra is classical.
Conditional	Application to a particular tree, a uniform complex family, a changed test bank, or an approximate metric requires the stated continuum bounds.
Executed	Symbolic seam, residual, finite-rank, resonance, scaling, and scalar-model checks recorded in Section 11.
Research	Validated geometric constants, certified modal truncation, and extension of the admissibility class are proposed work, not results.

The catalogue's L5 is a screening measure of relevance. It says nothing about historical novelty, truth of every argument in S, numerical speed, or the realized contribution. C2 is a separate, evidence-based assessment: the additions organize and quantify an important interface, but do not improve the headline conclusion or remove its hardest analytic assumptions.

2 The geometric statement and its normalization

2.1 The fixed tree and all quantifiers

Fix a finite rooted tree $\mathcal{T} = (V, E)$, fixed four-dimensional Ricci-flat vertex metrics h_v, quotient charts and orthogonal end identifications. Each joined group is a nontrivial finite subgroup of $O(4)$ acting freely on S^3. On exterior ends set $\tau = \log r$; on punctured ends set $\tau = -\log r$. Joined ends have

$$h_v - \delta = O(e^{-\mu\tau}), \qquad \mu > 0,$$

with scaled derivative bounds; the unjoined root exterior has $h_{\mathrm{root}} - \delta = O(r^{-q})$, $q > 1$. The hypotheses retain uniform metric estimates through order 20.

For a parent p, child c, and edge e,

$$x_p = e^{-2L_e} x_c, \quad a_c = a_p e^{-2L_e}, \quad a_{\mathrm{root}} = 1, \quad \rho = a_v r, \quad \tau_p + \tau_c = 2L_e. \tag{2.1}$$

At the seam, $\rho = a_p e^{-L_e} = a_c e^{L_e}$. The physical metric is $a_v^2 h_v$ in vertex coordinates. In particular an edge is not a unit-length adjacency in a graph Laplacian.

S's theorem concerns any sequence g_j converging on compact subsets of all open vertices with all $L_e^{(j)} \to \infty$, not necessarily at comparable rates. Let $M_j \to 0$ and require, through ten scaled derivatives,

$$|\rho^2 \operatorname{Ric}(g_j)|_{10} \le C M_j \begin{cases} e^{-\mu\tau} & \text{on a joined half-end,} \\ r^{-q} & \text{on the root exterior,} \\ 1 & \text{on a core.} \end{cases} \tag{2.2}$$

Constants may depend on this fixed tree and its charts. S asserts $E(g_j) = o(M_j)$; when $M_j = 0$, it asserts $E(g_j) = 0$. It does not assert a power exponent uniform over trees, a uniform number of vertices, an extension to trivial joined groups, or a Ricci-flow continuation theorem.

2.2 Renormalization and the degree-two scale

The functional is the combined limit

$$E(g) = \lim_{D\to\infty} \left[\int_{r<D} R(g)\, d\mu_g - \int_{r=D} (\partial_j g_{ij} - \partial_i g_{jj}) n_i \, dS_\delta \right]. \tag{2.3}$$

The integrals need not converge separately. With $g = \delta + p$, the scalar density is its linear Euclidean divergence plus

$$O(|p||\partial^2 p| + |\partial p|^2) = O(r^{-2q-2}).$$

The radial integral is $\int r^{1-2q} dr$, finite exactly when $q > 1$ for this bound. The variation of the linear flux cancels the leading boundary term; the remaining boundary products are $O(D^{2-2q})$. Consequently S obtains

$$DE_g(v) = \int \left\langle \tfrac{1}{2} R(g) g - \operatorname{Ric}(g), v \right\rangle_g d\mu_g. \tag{2.4}$$

A length rescaling, with compensating dilation at the root, sends E to $s^2 E$. Root displacements $O(r^{1-q'})$, $q' > 1$, yield the stated diffeomorphism invariance by the contracted Bianchi identity and vanishing boundary terms.

Differentiate the scale law. On a parent half the pairing with (2.2) is bounded by

$$C M a_p^2 \int_{\tau_0}^{L_e} e^{-(2+\mu)\tau} d\tau.$$

On the child half it is bounded by

$$C M a_p^2 e^{-4L_e} \int_{\tau_0}^{L_e} e^{(2-\mu)\tau} d\tau \le C M a_p^2 (1 + L_e) e^{-\min(2+\mu, 4) L_e}.$$

The finite core sum is bounded since $a_v \le 1$, and the root contribution is integrable. Thus $|E| \le CM$, and $E = 0$ for a Ricci-flat metric. This is S's preliminary estimate, not its strict little-oh improvement.

3 What the augmented inverse actually inverts

3.1 Weights, domain, and finite data

Retain S's choices

$$\begin{aligned} q_0 &= 1 + \tfrac{1}{2}(\min(q,2) - 1), & q_1 &= \tfrac{1}{2}(1 + q_0), \\ \lambda &= 1/n < \tfrac{1}{100}\min(\mu, 1), & \nu &= \lambda/4, \quad \nu' = \lambda/2, \quad \gamma_g = 2\lambda. \end{aligned} \tag{3.1}$$

In particular $1 < q_1 < q_0 < \min(q, 2)$ and $q_0 + q_1 > 2$. Use cylindrical measure $d\tau\, d\theta$. The metric domain X^k_γ, $k = 8$, includes core H^k, the norms of $e^{\gamma\tau}U$ on every joined half, and $r^{q_0}U$ on the root. For vectors the component is V/r. The source space has two fewer derivatives and multiplies the equation by the local squared length. These are weighted Sobolev operator spaces, not a finite list of jets. Orders 6 through 9 and the extra vector order are retained when traces or shape derivatives demand them.

A background b_H agrees with the fixed vertex metric to $\tau = H$ and is flat after $H + 1$, also on the root tail after a fixed truncation. The metric equation is

$$P_b(g) = \mathrm{Ric}(g) - \tfrac{1}{2}\mathcal{L}_{W_b(g)}g, \qquad W_b(g)^i = g^{jk}(\Gamma(g)^i_{jk} - \Gamma(b)^i_{jk}). \tag{3.2}$$

Its principal part is $-\frac{1}{2}g^{ab}\partial_a\partial_b g_{ij}$; the vector operator is the linearized tension of the identity. On flat ends they reduce to component Laplacians up to nonzero factors.

For a subset $D \subseteq E$ of opened edges, write the complete linear operator schematically as

$$\mathcal{A}_D(U, d) = (\mathcal{L}_D U - \Phi_D d,\ T_D U,\ J_{0,D}U,\ J_{1,D}U,\ \Pi_D U). \tag{3.3}$$

Here d is a finite source coefficient vector; T_D comprises core tests; J_0, J_1 are the full value and opposite-normal derivative jumps on $E \setminus D$; Π_D is the critical-channel parent trace. The last three rows are omitted on opened edges, while the opened ends retain their decay or explicitly prescribed asymptotic domain. The unknown is piecewise H^k before matching. Jump rows take values in infinite-dimensional trace spaces. Only the critical tests and compact augmentations are finite.

For real lengths the jump/test norms on edge e are

$$e^{\gamma L_e}H^{k-1/2}, \qquad e^{\gamma L_e}H^{k-3/2}, \qquad e^{\gamma L_e}(\text{finite critical space}).$$

The precise data product norm may be a fixed sum or maximum; all constants below must use the same declared choice. We use sum norms for residual ledgers. Finite spaces carry fixed Euclidean norms.

3.2 The all-mode Fredholm argument

For scalar Cartesian harmonic components of degree ℓ, r^ℓ and $r^{-\ell-2}$ give the following roots for $e^{\kappa\tau}$:

Component	Puncture	Exterior
Metric	$-\ell,\ \ell+2$	$\ell,\ -\ell-2$
Vector V/r	$1-\ell,\ \ell+3$	$\ell-1,\ -\ell-3$

Only equivariant degree spaces are present. The joined vector degree-zero channel vanishes: an invariant constant vector would be a fixed unit vector for any nonidentity group element, contrary to free action on S^3.

The weighted line is nonexceptional. For

$$(\partial_\tau - \kappa_1)(\partial_\tau - \kappa_2),$$

choose each first-order inverse in the direction in which its weighted exponential decays. The root separation is $2(\ell+1)$. Outside finitely many low channels the resulting kernel obeys

$$|K_\ell(\tau,\tau')| \le \frac{C}{1+\ell} e^{-c(1+\ell)|\tau-\tau'|}. \tag{3.4}$$

Its L^1 bound is $O((1+\ell)^{-2})$. One differentiated kernel uses one factor $1+\ell$; the equation supplies the second derivative. Squared harmonic summation yields a two-derivative Sobolev gain, including all high modes and their multiplicities. Finitely many remaining channels use their positive weighted gaps.

S then patches these end inverses to compact-core elliptic parametrices. Commutator errors gain one derivative on a compact set and are compact by Rellich; the far coefficient error is small in operator norm. Left and right parametrices give Fredholmness. Finite tests separate the kernel, and finitely many compactly supported sources represent the cokernel. The core can be enlarged to contain them. This yields a disconnected isomorphism. Stability when H grows uses a small operator perturbation of the already established inverse.

This sequence of arguments is essential. No determinant of a retained mode matrix proves compactness of the parametrix errors, closed range on the noncompact manifold, density of the selected sources in the relevant cokernel, or uniform high-mode bounds. The frame statement in Core-5 retains only its declared model space; it cannot replace (3.4).

3.3 Seam algebra and the missing critical datum

In a noncritical channel the incoming roots are

$$(\alpha,\beta) = (\ell+2,\ell) \quad \text{or} \quad (\ell+3,\ell-1).$$

Seam-normalized amplitudes A, B produce $j_0 = A - B$, $j_1 = \alpha A + \beta B$. Hence

$$\begin{pmatrix} A \\ B \end{pmatrix} = \frac{1}{\alpha+\beta} \begin{pmatrix} \beta & 1 \\ -\alpha & 1 \end{pmatrix} \begin{pmatrix} j_0 \\ j_1 \end{pmatrix}, \qquad \alpha+\beta = 2(\ell+1). \tag{3.5}$$

The derivative datum has one lower angular regularity, exactly offset by the denominator. For the metric degree-zero and vector degree-one critical channels there are three amplitudes:

$$(j_0, j_1, t) = (A + B - C,\ \kappa B,\ A + B), \quad \kappa = 2 \text{ or } 4. \tag{3.6}$$

Thus

$$B = j_1/\kappa, \quad A = t - j_1/\kappa, \quad C = t - j_0.$$

The critical determinant is κ. Without t, the vector $(1,0,1)$ is invisible. This is a genuine missing mode, not a small perturbation.

Cut incoming waves off before the core and correct their compact equation error using the disconnected augmented inverse. A constant wave with unit weighted seam amplitude has core size $O(e^{-\gamma L_e})$; growing waves are smaller there. Outgoing tails at any receiving seam have a strict decay gap beyond γ. S's full trace map is the direct sum of (3.5)–(3.6) plus a vanishing operator perturbation. This uses no comparison between different lengths. Injectivity follows by subtracting all corrected incoming waves and applying the disconnected inverse to the decaying remainder.

Opening an edge deletes that edge's incoming matching construction, not its remaining ends. There are $2^{|E|}$ partial disconnections, a finite number for this tree. Uniformity is obtained only after estimates have been established for each, taking the largest threshold and smallest allowed parameter radius. It is not a uniform theorem for growing trees.

4 Locally proved operator residual criteria

4.1 Two-sided defects on the full spaces

Theorem 4.1 (Full augmented residual criterion)**.** *For every $D \subseteq E$ and parameter ζ in a specified domain, let $\mathcal{A}_D(\zeta) : \mathcal{X}_D \to \mathcal{Y}_D$ and $B_D(\zeta) : \mathcal{Y}_D \to \mathcal{X}_D$ be bounded linear maps between Banach spaces. These spaces include all rows of (3.3). Suppose uniformly*

$$\|I - B_D\mathcal{A}_D\| \le q_L < 1, \qquad \|I - \mathcal{A}_D B_D\| \le q_R < 1, \qquad \|B_D\| \le b. \tag{4.1}$$

Then $\mathcal{A}_D$ and B_D are isomorphisms, and

$$\left\|\mathcal{A}_D^{-1}\right\| \le C_* := \min\left\{\frac{b}{1-q_L}, \frac{b}{1-q_R}\right\}, \qquad \|x - \widehat{x}\| \le C_* \left\|f - \mathcal{A}_D\widehat{x}\right\|. \tag{4.2}$$

All statements are on the displayed spaces, uniformly over the stated family.

Proof. Both products in (4.1) are invertible by norm-convergent geometric series. The first gives a left inverse $(B_D\mathcal{A}_D)^{-1}B_D$ of $\mathcal{A}_D$; the second gives a right inverse $B_D(\mathcal{A}_D B_D)^{-1}$. A left and a right inverse agree. The two estimates follow from the corresponding series. The same product argument makes B_D invertible. Apply the inverse to the exact residual. □

For a candidate $(\widehat{U}, \widehat{d})$, the residual norm in this theorem is, explicitly,

$$\begin{aligned}\delta = {} & \left\|f - \mathcal{L}_D\widehat{U} + \Phi_D\widehat{d}\right\|_{Y_\gamma^{k-2}} + |t_{\mathrm{core}} - T_D\widehat{U}| \\ & + \sum_{e \notin D} e^{\gamma L_e}\left(\left\|j_{0,e} - J_{0,e}\widehat{U}\right\|_{H^{k-1/2}} + \left\|j_{1,e} - J_{1,e}\widehat{U}\right\|_{H^{k-3/2}} + |t_e - \Pi_e\widehat{U}|\right).\end{aligned} \tag{4.3}$$

On complex contours replace these factors and the source norm by Section 6's strengthened norms. An evaluation error must be added to δ in those same units. One may not replace it by a Euclidean norm of sampled bulk values.

Neither two-sided smallness nor an inverse is proved by testing finitely many vectors. The conditions are operator norm bounds, including complement control. If one already imports S's isomorphism, a one-sided parametrix error suffices for a bound; the two-sided formulation above deliberately avoids silently assuming surjectivity from injectivity.

4.2 A finite block with an infinite complement

Apply (3.5)–(3.6) first to normalize the full trace map to $I + K$. Split its trace sequence space into low equivariant degrees and all degrees above N, with bounded orthogonal projections in the chosen Hilbert trace norm:

$$\mathcal{S} = \begin{pmatrix} A & B \\ C & D_\infty \end{pmatrix}, \qquad D_\infty = I + K_{hh}.$$

The low space is finite for fixed tree and N; the high space is not. Assume $\|K_{hh}\| \le d < 1$, so D_∞^{-1} exists and $\|D_\infty^{-1}\| \le (1-d)^{-1}$. The exact low Schur operator and load are

$$S_N = A - BD_\infty^{-1}C, \qquad g_N = g_l - BD_\infty^{-1}g_h. \tag{4.4}$$

An inverse for S_N, together with the high inverse, gives the full solution

$$x_l = S_N^{-1}g_N, \qquad x_h = D_\infty^{-1}(g_h - Cx_l).$$

This is an infinite-dimensional elimination with a finite reduced operator, not elimination of the tail by omission.

Proposition 4.2 (Transported trace residual)**.** *For any candidate x_l, x_h, set*

$$r_h = g_h - Cx_l - D_\infty x_h, \qquad r_N = g_N - S_N x_l.$$

Its full low residual is

$$r_l = r_N + BD_\infty^{-1}r_h. \tag{4.5}$$

If $\left\|S_N^{-1}\right\| \le s$, the exact errors obey

$$\|e_l\| \le s(\|r_l\| + \|B\|\,(1-d)^{-1}\,\|r_h\|), \qquad \|e_h\| \le (1-d)^{-1}(\|r_h\| + \|C\|\,\|e_l\|).$$

Proof. Solve the definition of r_h for x_h and substitute in $g_l - Ax_l - Bx_h$. This gives (4.5). Apply the same elimination to $\mathcal{S}e = (r_l, r_h)$ to obtain the bounds. □

The plus sign in (4.5) is important. It is the continuum trace counterpart of the private Schur residual formula. A reduced residual alone is insufficient if the high inverse action is approximate.

For an explicit analytic tail estimate, suppose the normalized mode-transfer bounds imply

$$\|K_{hh}\| \le C\sum_{\ell>N}(1+\ell)^r e^{-(\ell+1)t}, \qquad t = \min_{e\notin D}(\operatorname{Re} L_e - H) > 0,$$

where $r \ge 0$ absorbs the actual angular multiplicity and derivative factors. For $t \ge t_0 > 0$, direct comparison gives

$$\|K_{hh}\| \le Ce^{-(N+2)t/2}\sum_{\ell=0}^{\infty}(1+\ell)^r e^{-(\ell+1)t_0/2}. \tag{4.6}$$

Indeed split the exponential into two equal factors and bound the first by its value at $\ell = N+1$; enlarge the second sum. This proof includes all omitted modes. The constants C, r, t_0 must be derived for the actual trace construction and quotient representations. No such numerical constants are certified here. For an entirely opened tree there are no finite seam blocks, so this tail test is vacuous, but the disconnected bulk inverse remains mandatory.

If an approximate finite Schur matrix $\widetilde{S}_N$ has scaled smallest singular value s_0, and $\left\|S_N - \widetilde{S}_N\right\| \le \epsilon < s_0$, then $\left\|S_N^{-1}\right\| \le (s_0-\epsilon)^{-1}$. This follows by the reverse triangle inequality, including all inverse-action and rounding errors in ϵ. It is an absolute margin, not just a relative condition number.

5 Finite-rank changes that are genuinely finite-rank

Theorem 5.1 (Augmentation update on a fixed domain)**.** *Let $\mathcal{A} : \mathcal{X} \to \mathcal{Y}$ be an established bounded isomorphism, $G = \mathcal{A}^{-1}$. Let $U : \mathbb{C}^r \to \mathcal{Y}$ and $V : \mathcal{X} \to \mathbb{C}^r$ be bounded, complex-linear maps on the same fixed spaces. Put $T = I_r + VGU$. Then $\mathcal{A}' = \mathcal{A} + UV$ is invertible if and only if T is invertible, and in that case*

$$G' = G - GUT^{-1}VG, \qquad \|G'\| \le \|G\| + \|GU\| \left\|T^{-1}\right\| \|VG\| . \tag{5.1}$$

Proof. If T is invertible, multiplication of $(I+GUV)(I-GUT^{-1}V)$ gives I; the reverse multiplication also gives I, and (5.1) follows. Conversely, if $Ta = 0$, then $(I+GUV)GUa = GUTa = 0$. Invertibility of $\mathcal{A}'$ forces $GUa = 0$, and then $Ta = a = 0$. Thus T is injective and hence invertible in finite dimension. The norm bound is the triangle inequality. □

Here V, not a parameter-dependent conjugate transpose, is the analytic factor. In a real fixed matrix test one may write $V = V_0^*$; when parameters are complex, conjugating a holomorphic matrix would generally destroy holomorphy.

Replacing finitely many core tests, or changing the finite list of source columns while keeping its size, changes (3.3) by finite rank. For example a new test vector differs by a bounded map $\Delta T : \mathcal{X} \to \mathbb{C}^b$, followed by injection into the finite test rows. A new source column is a map from the finite coefficient space into the bulk source space. The theorem then applies after a fixed common coordinate identification and explicit norm conversion.

Changing an entire metric coefficient, moving a seam, changing an edge length, opening an edge, or changing quotient equivariance is not automatically finite rank. Matching rows are infinite-dimensional and the domain may change. Partial disconnections require separately constructed inverses. A low-rank update is not a substitute for S's uniform seam or contour argument.

For approximate old inverse actions $\widetilde{Z} \approx GU$, $\widetilde{x}_0 \approx Gf$, define

$$E_U = U - \mathcal{A}\widetilde{Z}, \quad r_0 = f - \mathcal{A}\widetilde{x}_0, \quad \widetilde{T} = I + V\widetilde{Z}.$$

With $C \ge \|G\|$,

$$\left\|T - \widetilde{T}\right\| \le \|V\|\, C\, \|E_U\| , \qquad \sigma_{\min}(T) \ge \sigma_{\min}(\widetilde{T}) - \|V\|\, C\, \|E_U\| . \tag{5.2}$$

If $s = V\widetilde{x}_0 - \widetilde{T}\widehat{t}$ and $\widehat{x} = \widetilde{x}_0 - \widetilde{Z}\widehat{t}$, expansion gives

$$f - \mathcal{A}'\widehat{x} = r_0 - E_U\widehat{t} - Us. \tag{5.3}$$

Thus the corrected field still needs the full bulk, test, and seam residual. Even $T = \varepsilon$ has condition number one for every nonzero scalar ε, while its inverse gain diverges. The absolute middle margin and (5.1)'s full gain are indispensable.

When $\mathcal{A}, U, V$ are holomorphic in local fixed-space charts and T^{-1} has a uniform bound, (5.1) is holomorphic there. A uniform nonvanishing determinant on a compact finite parameter set is useful, but determinant nonzero pointwise on a noncompact length domain alone gives no uniform inverse bound. This is a conditional analytic-parameter extension, with no claim that it alone preserves all-order length admissibility.

6 Complex lengths and nonlinear radii

6.1 What the contour estimate must pay

S continues each length to

$$Q(A, c_0) = \{L : \operatorname{Re} L > A,\ |\operatorname{Im} L| < c_0(\operatorname{Re} L)^2\}.$$

For $\ell = \operatorname{Re} L$, its contour is $\tau(u) = u + i(\operatorname{Im} L)\chi(u/\ell)$, real before $\ell/3$, ending at L, with $\tau' = 1$ near the seam. Bends lie in the exactly flat region. For real roots κ,

$$|e^{\kappa(\tau(u)-\tau(v))}| = e^{\kappa(u-v)}.$$

This preserves exponential decay, but not the measure or derivative constants: $|\tau'| \le C(1+\ell)$, $|\tau^{(j)}| \le C_j(1+\ell)^2$, and $|\tau'| \ge 1$.

The source norm is strengthened to

$$\|f\|_{Y_\gamma} + (1+\ell)^P \|\chi_\ell f\|_{Y_\gamma}, \tag{6.1}$$

with $\chi_\ell = 0$ before $\ell/6$, equal to one after $\ell/5$; seam norms also carry $(1+\ell)^P$. The fixed integer P pays all operations at the finitely many selected Sobolev orders, not an arbitrarily increasing expansion order.

Split $f = (1-\chi_\ell)f + \chi_\ell f$. The first summand is supported before $\ell/5$; propagation to the bend at $\ell/3$ gains $e^{-\delta_0(2\ell/15)}$ from the weighted root gap. The second is paid by (6.1). Far coefficient perturbations $O(e^{-\sigma u})$ contribute a bound proportional to $(1+\ell)^P e^{-\sigma\ell/5}$. A quadratic nonlinear term supplies a surplus $e^{-\nu\ell/6}$. Thus one needs genuine strict exponential margins, not an assertion that complex length has the same norm as real length.

For use in an explicit constant ledger define

$$\Phi_{P,s}(A) = \sup_{x \ge A}(1+x)^P e^{-sx} = (1+x_*)^P e^{-sx_*}, \quad x_* = \max\{A, P/s - 1\}, \tag{6.2}$$

where $P \ge 0, s > 0, A \ge 0$. Differentiate $P\log(1+x) - sx$ to prove the formula. This gives a concrete threshold test once P, s and their prefactors are known. S specifies how P can be chosen but does not supply a numerical P and constants for an actual tree. Our scalar tests do not supply them either.

6.2 A quantitative common nonlinear ball

Theorem 6.1 (Uniform radii and analytic parameters)**.** *For every $D \subseteq E$, let $\mathcal{F}_D(\zeta, \cdot) : \mathcal{X}_D \to \mathcal{Y}_D$ be continuously differentiable on a neighborhood of a closed radius-r ball about $c_D(\zeta)$. Let $B_D : \mathcal{Y}_D \to \mathcal{X}_D$ be a bounded linear isomorphism. Suppose, uniformly in all permitted parameters and D,*

$$\begin{aligned} \|B_D\mathcal{F}_D(c_D)\| &\le \eta, \\ \|I - B_D\mathcal{F}_D'(c_D)\| &\le \kappa, \\ \|B_D(\mathcal{F}_D'(u) - \mathcal{F}_D'(v))\| &\le L\|u - v\|. \end{aligned} \tag{6.3}$$

If

$$q = \kappa + Lr < 1, \qquad \eta + \kappa r + \tfrac{1}{2}Lr^2 \le r, \tag{6.4}$$

there is exactly one root in each ball, at distance at most $\eta/(1-q)$ from its center. Its derivative inverse has norm at most $\|B_D\|/(1-q)$; a uniform inverse bound also requires a uniform bound

on $\|B_D\|$. If the equations are jointly holomorphic in state and parameters, the centers and preconditioners are holomorphic in parameters in compatible local fixed-space charts, and the bounds hold on a common complex parameter domain, the roots are holomorphic there. Local roots agree on overlaps when the same uniqueness ball is used.

Proof. Set $T(u) = u - B_D\mathcal{F}_D(u)$. Along segments in the ball its derivative has norm at most q. Along a radial segment from c_D, the sharper bound is $\kappa + Lt$, which integrates to $\|T(u) - c_D\| \le \eta + \kappa r + Lr^2/2 \le r$. Banach contraction gives a unique fixed point; injectivity of B_D makes it a zero of the full equation. Subtracting the center gives the error bound. Since $B_D\mathcal{F}_D'(u_*)$ differs from I by at most q, its Neumann inverse gives the derivative estimate. In holomorphic charts, Picard iterates are holomorphic and converge locally uniformly, so their limit is holomorphic. Uniqueness gives the overlap statement. □

The factor $1/2$ in the self-map inequality uses the integrated derivative Lipschitz bound. It improves the more conservative $\eta + qr \le r$ test, but this is classical local analysis, already present in PDE-first Report 8, not a new geometric theorem.

For S's branch the equation and parameters are

$$P_b(h_*) + \tfrac{1}{2}\mathcal{L}_{m\psi}h_* = d\varphi, \qquad h_* = b + U, \qquad \pi U|_e = e^{-\lambda L_e} z_e.$$

The ordinary vector z collects core values, m, and critical seam coefficients. Equation-source scalings are not displacement scalings: physical vector sources have components $a_v^{-1}\psi_v$, physical covariant source components are $a_v^{-2}\varphi_v$, whereas a displacement scales as $a_v V_v$. These conventions make normalized core equations independent of lengths.

Apply Theorem 6.1 only after estimates in (6.1). On a radius-δ ball, the Lipschitz constant of the preconditioned quadratic *remainder map* contains a factor bounded by

$$C\delta(1 + \Phi_{P,\nu/6}(A)),$$

whereas the theorem's *derivative-Lipschitz* constant L has a bound of order $C(1 + \Phi_{P,\nu/6}(A))$, without that radius factor. The small radius enters its contraction bound through Lr. Substituting the remainder-map constant for L would incorrectly count smallness twice. Seam forcing contains $\Phi_{P,\lambda-\nu}(A)|z_e|$. S's ordering is: weights and core inverse bound; a small nonlinear radius; truncation H; ordinary parameter radius; finally a large length threshold $A > 10(H+1)$. The background coefficient and inverse-matrix estimates are uniform in H, which prevents circular dependence. The same fixed choices must work at ν, ν' and on every partially disconnected tree.

6.3 Holomorphy is more than real-part estimates

The contours themselves depend smoothly on $\operatorname{Re} L, \operatorname{Im} L$, not holomorphically. S uses local endpoint charts and a shape derivative to identify different contours. If $v = \delta\tau$ fixes endpoints, then

$$\delta\partial_\tau = -(\partial_\tau v)\partial_\tau, \qquad \delta(\partial_\tau U) = \partial_\tau(v\partial_\tau U) - (\partial_\tau v)\partial_\tau U = v\partial_\tau^2 U.$$

The proposed field variation $v\partial_\tau U$ lies at one lower Sobolev order. Uniqueness there, with the same tests and appropriately lowered trace orders, identifies it as the actual variation. Fixed endpoint matching and core data are unchanged. Homotopy through increasing-real-part contours then identifies local continuations and removes monodromy. An opened-end bend is returned to the real ray and handled by compactly supported deformation.

An implementation of the residual theorem does not prove this contour-independence step. Likewise pointwise matrix analyticity is not a substitute for a holomorphic family of full operators on locally identified Banach spaces.

7 Tails, finite length expansions, and common domains

7.1 Triangular tail extraction without a formal inverse

On an opened joined end S extracts, to every nonintegral cutoff B,

$$Y = \sum_{l<B} e^{-l\tau} \sum_{j=0}^{d_l} \tau^j Y_{lj}(\theta) + Y_{>B}, \qquad l \in \mathbb{Z},$$

with finite harmonic coefficients, a finite lower growth bound, and remainder at any fixed weight strictly below B. Apply the flat operator to $e^{-l\tau}p(\tau)$. Away from roots the polynomial equation is triangular with a nonzero diagonal. At a root, integrate one polynomial factor, raising its degree by one; choose zero primitive constant and retain the omitted constant as a free homogeneous coefficient. Roots are distinct, so there is no repeated-root ambiguity.

For example, for a scalar exterior degree-zero channel $P = \partial_\tau(\partial_\tau + 2)$,

$$P(-\tfrac{1}{2}\tau e^{-2\tau}) = e^{-2\tau}.$$

This resonant logarithm cannot be represented by a single pure exponential. After subtracting forced primitives, root kernels at a noncritical weight improve the remainder; homogeneous coefficients at crossed roots are recorded.

The small nonlinear solution initially has positive decay. Its nonlinearity is quadratic, so its source has twice a slightly weaker rate. Each successive Taylor remainder containing the previous remainder also contains a fixed positive-decay factor. Iteration gains a fixed positive rate each time; integer roots and addition of degrees give integer tails. Polynomial angular coefficients preserve finite harmonic support at each finite order. This is not a uniform angular analyticity estimate over all orders.

For an auxiliary field with finite prescribed nondecaying data, write

$$Y = \zeta T + \mathcal{A}_U^{-1}(f - \mathcal{A}_U(\zeta T)), \tag{7.1}$$

where T cancels every nondecaying forced term and fixes all free incoming constants until the residual is in the original decaying source space. Coefficients of $\mathcal{A}_U - \mathcal{P}$ have positive integer decay, making the recursion triangular. The inverse in (7.1) is the same basic inverse as for the small metric; the large auxiliary field does not enter a new small nonlinear ball.

This distinction is the mechanism preserving a common parameter domain. It also explains why simply increasing a global weighted inverse order after every coefficient is unnecessary. Appended systems are solved in triangular order with $\mathcal{A}_U$ on every diagonal; on still-finite halves their sources have at least two decaying factors to pay complex contour strengthening. At newly opened ends the earlier fields start decaying. On older opened ends finite growth is retained explicitly.

7.2 Usable order and unequal lengths

Open an edge of length L and let Y_0 solve the resulting system. A correction has form $e^{-pL}L^jY_{pj}$. If Y_{pj} grows at most $e^{b\tau}$ on the new pair of ends, define

$$s = p - \max(0, b) > 0.$$

All tags lie on $\lambda\mathbb{Z}$. Products preserve a lower bound by adding tags, because $\max(0, b_1 + b_2) \le \max(0, b_1) + \max(0, b_2)$. At each order solve the full forced bulk coefficient first, with zero free

incoming constants; then cancel the leading seam residual using (3.5)–(3.6). A mode $e^{b\tau}$ with leading seam order s is multiplied by $e^{-(s+b)L}$. Its lower tails have later orders.

Restricting $\operatorname{Re}\tau \le \operatorname{Re} L$ gives an unweighted bound

$$C(1+\operatorname{Re} L)^C e^{-s\operatorname{Re} L}.$$

The basic weight costs $e^{\nu \operatorname{Re} L}$ once after the product estimate, not once per factor. Polynomial strengthening uses a strict additional exponential surplus. Other finite halves retain decay and the two-factor surplus in their own lengths. Older opened ends retain the full factor e^{-pL} in a sufficiently high finite tail norm. These separate estimates are needed when lengths diverge at unrelated rates.

7.3 Finite parts and the common analytic domain

For an angularly integrated density $f(\tau)$, let T be its finite nondecaying tail and $Q_{\alpha j}$ fixed primitives of $e^{\alpha\tau}\tau^j$, $\alpha \ge 0$. Then

$$\int_{\tau_0}^{L} f = C_f + \sum_{\alpha,j} c_{\alpha j} Q_{\alpha j}(L) - \int_L^\infty (f-T).$$

The constant C_f includes the compact cutoff correction; ignoring it changes coefficients. A positive remainder rate makes the last integral continuous by weighted Cauchy–Schwarz. Locally uniform dominated integration makes it holomorphic. Complex integration uses $d\tau$ and contour independence.

Ricci-square volume is scale neutral in dimension four; scalar-volume density carries $a_v^2 e^{\pm 2\tau}$, including e^{-4L} on the newly opened child side. To obtain scalar remainder order B, S takes tag cutoff strictly above $B+\nu+2+\delta$ and then enough tail precision for all finite growth. The allowances pay the basic weight, worst radial factor, and all polynomial costs.

For every partial disconnection fix the same small branch and inverse on a common smaller domain

$$\operatorname{Re} L_e > A_*, \quad |\operatorname{Im} L_e| < c_*, \qquad |z - z_*| < \rho_*. \tag{7.2}$$

Every coefficient of every order is defined there by finite linear solves, primitives, tail limits and integrals. Estimates, tail-start thresholds and remainder quadratic regions may depend on order, but the analytic domain (7.2) does not. An intersection of infinitely many shrinking polydiscs would not suffice.

8 The observable and the path-selection dependency

8.1 Comparison with the actual metric

S first gauges the actual real sequence by solving the augmented tension equation, giving $W_b(g) = m\psi$. Small V/r in scaled C^1, properness, and degree one make the displacement isotopy a diffeomorphism. Its root decay is $V = O(r^{1-q_0})$, which is within the invariance class of (2.3). Choose the branch parameters from the actual core and seam tests. Weight $\gamma_g > \lambda$ makes the seam parameters small. The actual metric solves the branch equation with extra source $\operatorname{Ric}(g)$. Uniform inverse and nonlinear absorption give

$$\|g - h_*\|_{X^k_\nu} \le CM, \quad \|\operatorname{Ric}(h_*)\|_{Y^{k-2}_\nu} \le CM, \quad |E(g) - E(h_*)| \le CM^2. \tag{8.1}$$

The last estimate integrates (2.4): both the Ricci source and metric variation are $O(M)$. Its root integrability is $\int r^{1-2q_0} dr < \infty$.

Define

$$F = E(h_*), \qquad J = \int |\operatorname{Ric}(h_*)|^2 \, d\mu_{h_*} + \int_{\text{root},\, r>r_0} r^{2q_1} |\operatorname{Ric}(h_*)|_\delta^2 \, dx. \tag{8.2}$$

On real parameters $J \geq 0$ and $J \leq CM^2$ on the comparison sequence. On complex parameters the contractions are bilinear without conjugation. Their holomorphic continuations are not nonnegative energies. Root convergence uses $\int r^{2q_1-2q_0-1} dr < \infty$.

The first variation and the uniform inverse for ordinary derivatives give the dz part of

$$|dF| \leq C\sqrt{J} \left(\sum_\alpha |dz_\alpha| + \sum_e e^{-cL_e} |dL_e| \right). \tag{8.3}$$

On the root the dual integral is $\int r^{3-2q_0-2q_1} dr < \infty$. Differentiating a length produces seam data of size

$$C\{e^{-(\nu'-\nu)L_e} + e^{-(\lambda-\nu)L_e}\}|dL_e|.$$

Transporting the seam in a fixed logarithmic collar adds an L^2 variation $O(a_p^2 e^{-2L_e}|dL_e|)$. Descendant squared scales have derivative $-4a_v^2$; their total volume is at most $Ca_p^4 e^{-4L_e}$, since

$$a_c^4 \int^{L_e} e^{4\tau} d\tau \leq Ca_p^4 e^{-4L_e}.$$

Thus any fixed $0 < c < \min(\nu' - \nu, 2)$ is admissible after constants are adjusted.

8.2 The precise admissibility obligations

S's path theorem is stronger than ordinary analytic curve selection at a finite point. In each length, each initial function and each recursively extracted coefficient must be bounded and real holomorphic on products of quadratic regions, and admit

$$P \sim \sum_{d \in D_i} e^{-dL_i} \sum_{k=0}^{k_d} L_i^k P_{dk}(L_{\widehat{i}}, z).$$

The sets $D_i \subset [0, \infty)$ are locally finite; the zero exponent has no positive polynomial degree. At every cutoff B, the remainder must be $O(e^{-(B+\epsilon_B)\operatorname{Re} L_i})$, $\epsilon_B > 0$. All recursively extracted coefficient formulas are holomorphic on a common ordinary neighborhood and a common fixed-width band in remaining lengths. Grouping lengths $\operatorname{Re} L_i = c_i U + o(U)$, $c_i > 0$, must retain locally finite total weights and strict remainder gains.

S proves these for F, J using Section 7. Boundedness rules out negative exponents and nonconstant zero-degree polynomials. Uniqueness of finite expansions identifies coefficients from different cutoffs; common inverses and finite linear constructions give (7.2). Successive length expansions with a strict surplus give grouped expansions; finitely many positive-grid multi-indices lie below a fixed total weight.

8.3 Why the multivariable path proof cannot be replaced by a matrix test

The full path section of S has the following linked steps.

1. *Sector uniqueness.* On sectors with summable angular loss $\omega \geq (\log r)^{-2}$, bounded holomorphic functions superexponentially small on a central sector vanish. A half-plane map $z \exp A(z)$, with $A(z) = \sum a_n/(1 + z/2^n)$, pushes the domain into the sector. Harmonic measure of a central half-disc arc is $\gtrsim 1/r$; applying subharmonicity to an e^{-Br} bound and then sending $B \to \infty$ gives zero.

2. *Clock fields.* Begin with meromorphic Puiseux germs. Adjoin exponential expansions in successively faster clocks, with strictly better remainders. Leading monomials, inversion, differentiation and conjugation give a Hardy field. Change of clock must preserve sector access and the distinction $\operatorname{Re} x_{\text{slower}} = o(\operatorname{Re} x_{\text{faster}})$, not merely a real-ray comparison.

3. *Uniform parameter limits.* A holomorphic family with a locally uniform finite real-ray limit has the same limit on a suitable sector. This protects nonzero contour denominators and composition domains after specialization.

4. *Regular systems.* Scalar normalization, divisible positive sizes, finite analytic operations on recorded domains, and submersive path families are maintained simultaneously. Fixed-circle preparation uses power sums of roots and polynomial division. Root clusters and rescaling establish real closedness of the real scalar field.

5. *Independent reciprocal length.* If a fastest center X is not within bounded distance of a lower scalar, $p = 1/X$ has $|p - c| \gtrsim p^2$ for every real lower scalar c. Matched disc and unmatched annular charts normalize finite expressions. A finite path specialization preserves $pM' \to 0$, with the lower last clock bounded by $M' = 1 + \sum L_i'$. This prevents a circular choice of the fastest clock.

6. *Centers and zero expansions.* Write comparable lengths $L_i = c_i X + D_i$. Unbounded slower differences, with their signs, become lower centers. Common analytic domains and identity detection through submersive path families are needed when infinitely many coefficients vanish. One-variable uniqueness alone does not establish this multivariable step.

7. *Ordinary-variable elimination.* Weierstrass preparation reduces signs to polynomial signs. Hermite trace forms $\operatorname{Tr}_{\mathbb{R}[y]/(Q)}(Pvw)$ count signs at distinct real roots even with multiplicity. Roots of D and D', plus extreme intervals, test all real sign patterns. Reverse lifting by roots, midpoints and infinitesimals preserves the ordinary limits.

Finite preparation data, not infinitely many separate leading-coefficient conditions, are preserved on the chosen path. Undoing the linear changes and restoring bounded shifts puts the *actual* coordinates in the Hardy field. Their derivatives then have fixed eventual sign or vanish, giving eventual monotonicity. The path is analytic at sufficiently large finite times; no analytic extension to its infinite endpoint is claimed.

For the explicit example $e^{-L_1} - L_2 e^{-L_2} = 0$, choose $L_2 = e^u$, $L_1 = e^u - u$. Equal limiting ratios do not permit discarding the unbounded difference $-u$. Neither a finite Taylor jet at $e^{-L} = 0$ nor a finite list of modal moments preserves this phenomenon.

8.4 The final contradiction and a limited generalization

If the source inequality fails, $|E(g)| \geq \varepsilon M$ on a subsequence. From (8.1)–(8.2) one obtains $F \neq 0$, $F \to 0$, and $F^2 \geq c_0 J$. Strict parameter smallness gives an interior ordinary subsequential limit. S's admissibility and path theorem produce a path preserving all these conditions with monotone actual coordinates. Along it

$$|d \log |F|| \leq C c_0^{-1/2} \left(\sum_{\alpha} |dz_\alpha| + \sum_{e} e^{-cL_e} |dL_e| \right).$$

Each bounded monotone z_α has finite variation, and $\int e^{-cL_e} |dL_e| = c^{-1} e^{-cL_e(u_0)}$. Thus $\log |F|$ has finite total variation, contradicting its limit $-\infty$.

Proposition 8.1 (Integrable differential weights)**.** *Assume the same admissible path-selection conclusion for F, J, but replace e^{-cL_e} in (8.3) by any nonnegative measurable $w_e(L_e)$ with $\int_A^\infty w_e(s)\,ds < \infty$. The same contradiction follows.*

Proof. On a monotone length path, substitution bounds the weighted variation by $\int_{L_e(u_0)}^\infty w_e(s)ds$. All other steps are unchanged. □

For example $w_e(s) = (1+s)^{-p}$, $p > 1$, works in this final argument; the integral from A is $(1+A)^{1-p}/(p-1)$. For $p \le 1$ that integral diverges, so this variation bound alone no longer supplies the contradiction. This is not a counterexample to S's stronger admissible setting. The elementary observation does not show that an altered geometry satisfies such a differential estimate, nor that polynomially decaying scalar families satisfy S's exponential admissibility conditions.

9 A source-specific approximation tolerance

Proposition 9.1 (Perturbed comparison metrics)**.** *Assume the uniform inverse/trace setup and first-variation pairing of S on a real small family. Let $\widehat{h}$ be a genuine matched metric in the same root gauge and decay class, and suppose*

$$\left\|\widehat{h} - h_*\right\|_{X_\nu^k} \le \epsilon, \qquad \|\mathrm{Ric}(h_*)\|_{Y_\nu^{k-2}} \le CM.$$

Assume the segment joining these metrics stays in a ball where Ricci is uniformly Lipschitz into the source space and the first-variation pairing has a uniform product bound. Then

$$|E(\widehat{h}) - E(h_*)| \le C'(M\epsilon + \epsilon^2). \tag{9.1}$$

Consequently an established $E(h_*) = o(M)$ is preserved if $\epsilon = o(1)$ and $\epsilon^2 = o(M)$; in particular $\epsilon = O(M)$ suffices.

Proof. Write $v = \widehat{h} - h_*$ and $h_t = h_* + tv$. Lipschitz continuity gives $\|\mathrm{Ric}(h_t)\| \le CM + C_R t\epsilon$. The product pairing in (2.4), with the root and physical scale integrals already checked, is at most $C_P(CM + C_R t\epsilon)\epsilon$. Integrate $0 \le t \le 1$. Divide by $M > 0$ to obtain the stated sufficient conditions. □

If the candidate and exact solution belong to the same contraction ball, then

$$\|u_* - \widehat{u}\| \le \frac{\|B\mathcal{F}(\widehat{u})\|}{1-q} \le \frac{\|B\|}{1-q}\,\|\mathcal{F}(\widehat{u})\|\,.$$

Thus a nonlinear full residual certificate with uniform C_{inv} gives $\epsilon \le C_{\mathrm{inv}}\delta$, and a residual of order M preserves the comparison's $O(M^2)$ accuracy. A fixed absolute tolerance independent of $M \to 0$ does not. A residual merely tending to zero also does not guarantee $\epsilon^2 = o(M)$. Division by M concerns positive indices; a zero-M index requires the original exact-zero convention and zero approximation error, as $\epsilon = O(M)$ imposes.

The proposition cannot be applied directly to a discontinuous piecewise candidate with unresolved jump rows: (2.3) presupposes a genuine metric. First solve or rigorously lift the mismatch, account for the lifting in the full norm, and ensure positivity and root decay. The residual theorem may certify proximity of piecewise fields to a matched solution, but that alone does not license evaluating the smooth-metric functional on the discontinuous candidate.

10 Private-corpus transfer and exclusions

Core-5's quantified Neumann tail states $\|A^{-1} - B_K\| \le q^{K+1} \|D^{-1}\| / (1-q)$ for $A = D(I-M)$, $\|M\| \le q < 1$, with the order of factors retained. Section 4 uses its valid Banach-operator analogue only after the spaces and two-sided defect are specified. A spectral-radius estimate alone is inadequate for a prescribed nonnormal norm.

The same report's frame theorem requires a lower frame bound on the retained space. Its warning about component constants is directly analogous to retaining each critical seam datum and each kernel test, but the analogy is not an identification of a tensor elliptic operator with a graph Laplacian. A finite core test bank separates a finite-dimensional kernel only after Fredholmness is known. The Schur source and inexact update passages supply the exact signs in (4.5) and (5.3).

Nonlinear Report 1 provides a preconditioned residual ball, finite inverse perturbation estimates, and bounded coordinate transfer. If $C : \mathcal{X} \to Z$, $S : \mathcal{Y} \to W$ are bounded isomorphisms, then

$$\widehat{\mathcal{F}}(z) = S\mathcal{F}(C^{-1}z), \qquad \widehat{\mathcal{F}}' = S\mathcal{F}'C^{-1}, \qquad \widehat{B} = CBS^{-1}.$$

Conjugating $I - B\mathcal{F}'$ costs $\|C\| \|C^{-1}\|$ unless the coordinate norms are transported exactly. This prevents a well-conditioned moment matrix in arbitrary coordinates from implying a physical norm bound. A redundant frame must impose coefficient-range consistency; a truncated frame is not a bounded isomorphism on the full end space.

PDE-first Report 8's coercive inverse theorem assumes a bounded Lipschitz domain, H_0^1, and a positive Hermitian form. Its Green equivalence retains the boundary lifting and the operator realization. These hypotheses do not hold automatically for S's noncompact, augmented, potentially noncoercive tensor problem or its complex contours. Its heat equivalence assumes a positive self-adjoint realization and genuine initial data; no heat theorem is used to prove this static Ricci-flat estimate. The semilinear section distinguishes inverse equivalence from contraction, and its nonlinear certificate includes the self-map condition used in Theorem 6.1. The full quasilinear derivative must retain coefficient variations; frozen coefficients are a different iteration.

11 Executed checks and their limits

The companion local script `tests.py` was executed with Python 3.10.9 and SymPy 1.11.1 at 2026-10-08 02:16 UTC (October 7 locally). All 30 checks passed: 29 exact rational/symbolic checks and one separately marked floating-point scalar-model probe. It is not a PDE solver. The execution JSON records runtime version, pass/fail results and numeric values. The checks are independently reproducible:

1. Both seam determinants and inverse formulas, including the null vector when the critical test is removed.
2. A finite block example of the Schur residual with nonzero high residual, detecting the sign and omitted source error.
3. An invertible rational augmented model and rank-one update, checking both inverse products and the inexact full residual formula.
4. A small absolute middle denominator with condition number one, and a singular update rejected by the middle test.
5. The resonant primitive $P(-\tau e^{-2\tau}/2) = e^{-2\tau}$, the opposite-normal sign, and the descendant scale derivative.

6. Every partial-disconnection subset of a three-edge *synthetic comparison matrix*, with row-sum invertibility bounds checked. This checks bookkeeping across eight subsets, not the corresponding geometric operators.

7. The explicit supremum (6.2), a scalar nonlinear radius, and the multiscale path identity from Section 8.

For concreteness, the Schur example has high residual $49/60$ and low residual $(17/20, -13/10)$, so ignoring the transported term is detectable exactly. The nonsingular rank-one update has middle scalar $109/78$. The largest synthetic row-sum bound over the eight subsets is $2/25$. For $F(x) = x + x^2 - 1/100$, $B = 1$, center zero and radius $1/50$, the exact nonlinear bounds are $q = 1/25$ and self-map radius $13/1250 < 1/50$. For the illustrative contour surrogate $P = 4, s = 1/2, A = 12$, $\Phi_{P,s}(A)$ is approximately 70.7956409178. This last value deliberately does not imply smallness; the threshold still depends on the missing geometric prefactor and required margin.

The parent supplied a separate nine-check SymPy 1.11.1 batch for variable-degree metric/vector trace blocks, the variable-κ critical block and rank loss without the critical test. Its nine checks also passed. It is corroborating finite algebra, not an extra mathematical contribution and not a test of the infinite complement.

The independent source auditor read all 3208 lines of the eleven build inputs and reported no substantiated P1/P2 defect in that bounded pass. It supplied no finite mathematical batch or continuum numerical enclosure. The report integrates its important qualifications: the simultaneous clock/normalization induction, especially finite-data substitution with all-order coefficients, and the complex inverse/contour-homotopy construction remain high-risk, non-formally verified steps. No absence-of-defects certificate is inferred from that disposition.

A focused follow-up reviewed the new bounded-operator, nonlinear, integrable-weight, and observable proofs and found no substantiated P1/P2 defect under their hypotheses. Its P3 application-wording clarification has been incorporated: the remainder-map Lipschitz constant and the derivative-Lipschitz constant in the radii theorem are now explicitly distinct. Linearity, joint holomorphy, the uniform preconditioner norm, the common nonlinear ball and zero-M convention are also made explicit. This review does not upgrade the global-source disposition or the contribution score.

No finite test establishes (3.4) on an actual quotient end, the uniform compact-core inverse, the contour polynomial exponent, the full Banach nonlinear constants, or the all-order path-selection hypotheses. In particular no measured speedup, interval-arithmetic continuum certificate, or global source verification is reported.

12 Comparison, limitations, and next research steps

12.1 Public literature boundary

A public-title/URL check on October 7, 2026 located the pinned OpenAI source, Lockhart–McOwen's noncompact elliptic paper [5], Ozuch's desingularization paper [6], Speissegger's quasianalytic algebra paper [7], and the published Deruelle–Ozuch ALE functional paper [8]. These are primary sources. The check verifies relevant publication records and context, not an exhaustive novelty search or independent reading of every external proof.

Lockhart–McOwen is the weighted noncompact Fredholm background. Ozuch supplies close gluing/modulo-obstruction context. Deruelle–Ozuch treats an ALE functional involving a potential minimization near a fixed background; S's unminimized scalar-minus-flux functional and independently degenerating tree are different. Speissegger supplies one-variable quasianalytic context; S

explicitly undertakes its multivariable specialization and selection argument. Neither these public precedents nor the private operator identities are claimed as newly discovered here.

12.2 Benefits and costs

The residual framework makes source, gauge, kernel, cokernel and seam errors visible in compatible norms. Exact low/high elimination exposes the required complement estimate. Finite-rank reuse can avoid reconstructing old inverse actions after a small test/source change, but only with a full gain estimate and stable factorization. The analytic radius separates residual smallness, derivative smallness and ball invariance. The observable bound supplies a meaningful tolerance as $M \to 0$.

The costs are substantial. Validated core bounds and high-mode constants may be harder than a fresh proof for a particular tree. Schur complements can be dense and poorly conditioned; no runtime advantage follows from algebra alone. The conservative sum norm can overestimate cancellations. Complex contour norms are stronger than real norms. Enumerating $2^{|E|}$ partial disconnections is finite but scales poorly. The all-order common-domain and multivariable path arguments remain outside finite numerical verification.

12.3 Research program with explicit acceptance conditions

First, choose an explicit geometric tree, quotient representations and charts, and derive a rigorous kernel/core parametrix bound at the selected weights. Numerical singular values must be complemented by estimates for all unretained bulk and angular modes.

Second, construct validated trace Schur actions using (4.6). Record each partial disconnection, coefficient truncation error, absolute finite margin, and the resulting full inverse bound. This would turn Section 4.2 from a conditional interface into an actual certificate for that tree.

Third, measure the cost of finite-rank source/test changes against reassembly and refactorization, including residual evaluation, modal tails, interval overhead and norm conversion. A speed claim would require that complete comparison.

Fourth, determine numerical P, s and prefactors for (6.1), then solve the explicit threshold and radius inequalities jointly in the noncircular order described above. Uniform analytic charts and contour homotopies must still be supplied.

Finally, seek a broader admissibility algebra compatible with integrable differential weights, or a quantitative modulus stronger than the sequential little-oh conclusion. Proposition 8.1 only relaxes the final variation step. Neither broader path selection nor a tree-uniform power law is proved or conjectured as already established.

A Read locators and reproducibility contract

All relative source paths below are under

```
_sources/openai-math/preprints/Path-selection-and-an-elliptic-inequality-on-d
egenerating-Ricci-flat-trees-September-24-2026/
```

The freeze manifest records the actual SHA256 of each read input, exact line intervals or labels, and the final output, execution, QA, and review records. This appendix gives human-readable locators; a path alone is not a proof of provenance. The seven section filenames below are in `build/sections/`.

Input	Actual read coverage
`01-introduction.tex`	Complete, lines 1–193; aims, precedents, reduction.
`02-tree.tex`	Complete, lines 1–156; geometry, renormalization, theorem.
`03-path.tex`	Complete, lines 1–1320; definitions and every path-selection proof.
`04-family.tex`	Complete, lines 1–572; all inverse, gauge, analytic-family and comparison proofs.
`05-tails.tex`	Complete, lines 1–213; finite-growth and triangular auxiliary equations.
`06-length.tex`	Complete, lines 1–166; tag induction and remainder estimates.
`07-observables.tex`	Complete, lines 1–307; finite parts, common domains, differential estimate, conclusion.
Main/support source files	`README.md`, `build/paper.tex`, `build/preamble.tex`, `build/references.bib`, `build/figures/tree-scales.tex`, complete.
Private Core-5	`RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`, lines 189–535: inverse, frame, Schur, update and residual passages.
Private nonlinear Report 1	`RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`, complete: perturbation, ball/radii, coordinate transfer and limits.
Private PDE-first Report 8	`RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`, lines 53–236, 270–423, 784–872: domains, coercive/Green/heat equivalence, nonlinear and interface passages, certificates.

B Build and review status

The native compiler was attempted and returned a platform-initialization failure: `Unable to find standard directories for platform`. The source-open request returned `queued`, which does not prove a visible editor tab opened. Neither outcome is represented as successful native compilation. The requested installed MiKTeX helper supplies the separately requested disk PDF. Parent review and the final disk-export/visual-QA results are recorded before freeze; their exact records and output hashes are in the manifest.

References

[1] OpenAI. *Path selection and an elliptic inequality on degenerating Ricci-flat trees*. September 24, 2026. Pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public source directory.

[2] Muralidhara Subbarao. Private Core-5 report, `TR-core-pat-5.tex`. Locators: `thm:neumann`, `thm:frame`, `prop:schurresidual`, `thm:woodbury`, and adjacent residual sections. Read locally; not uploaded.

[3] Muralidhara Subbarao. Private nonlinear Report 1, `stability.tex`. Locators: `thm:pert`, `thm:ball`, `eq:radii`, `prop:transfer`. Read locally; not uploaded.

[4] Muralidhara Subbarao. Private PDE-first Report 8, `TR8-lin-non-lin.tex`. Locators: `thm:coercive`, `eq:operator-domain`, `thm:green-equivalence`, `thm:heat-equivalence`, `sec:semilinear`, and the local a posteriori certificate. Read locally; not uploaded.

[5] R. B. Lockhart and R. C. McOwen. *Elliptic differential operators on noncompact manifolds*. Ann. Scuola Norm. Sup. Pisa (4) 12 (1985), 409–447. https://www.numdam.org/item/ASNSP_1985_4_12_3_409_0/.

[6] T. Ozuch. *Noncollapsed degeneration of Einstein 4-manifolds II*. Geometry & Topology 26 (2022), 1529–1634. https://arxiv.org/abs/1909.12960.

[7] P. Speissegger. *Quasianalytic Ilyashenko algebras*. Canadian J. Math. 70 (2018), 218–240. https://arxiv.org/abs/1606.02751.

[8] A. Deruelle and T. Ozuch. *A Lojasiewicz inequality for ALE metrics*. Ann. Sc. Norm. Sup. Pisa (5) 26 (2025), 1965–2042. https://journals.sns.it/index.php/annaliscienze/article/view/6142.